Live Prototype:
https://returlooptesting-khwd.vercel.app/

# Checkpoint 6 · Usability Testing & Iteration Matrix

Empirical Testing Across 6 Campus Users, Quantitative Metrics & 4 Core Engineering Fixes | INT335

## 1. Quantitative Usability Testing Protocol & Results

| Test # | Participant Name | User Profile | Task Evaluated | Duration | Task Success | Observed Usability Hitch |
|---|---|---|---|---|---|---|
| **T1** | **Rohan Verma** | Hosteller (2nd Yr CSE) | Report lost calculator with secret proof | 48s | Passed | Confused between public description and secret proof. |
| **T2** | **Aman Sharma** | Day Scholar (3rd Yr Mech) | Browse campus found feed on mobile | 30s | Passed | Filter dropdowns were too small on touch screen. |
| **T3** | **Officer Gurpreet** | Security Guard (Block 34) | Verify claim & release item | 65s | Passed (Aid) | Struggled with technical English terms; needed Punjabi. |
| **T4** | **Simranjeet Kaur** | Fresher Hosteller (GH-2) | Review match alert notification | 22s | Passed | Expressed fear that finder might see her phone number. |
| **T5** | **Staff Manjit Kumar** | Library Desk Supervisor | Log batch of 3 lost glasses | 95s | Passed | Requested fast duplicate item logging for exam weeks. |
| **T6** | **Sahil Gupta** | iPhone User (Mobile Chrome) | Change language in navigation bar | Failed | Failed | Language modal was cut off at top under Dynamic Island. |

## 2. Feedback-to-Iteration Matrix (The 4 Critical Engineering Fixes)

| Hitch Identified | Root Cause Diagnostics | Engineering Fix Implemented in Code | Empirical Re-Test Validation |
|---|---|---|---|
| **Fix 1: Mobile Modal Clipping** | iOS WebKit viewport safe-area inset clipped modal inside sticky navbar stacking context. | Refactored `LanguageSelectorModal.jsx` using React `createPortal` rendering to `document.body` with `env(safe-area-inset-top)` padding. | Sahil Gupta re-tested on iPhone Chrome; modal rendered centered with 100% visibility. |
| **Fix 2: Staff Language Barrier** | Security guards found English technical terms intimidating, causing slow claim handovers. | Implemented complete 7 Indian regional languages localization (Hindi, Punjabi, Telugu, Tamil, Bengali, Marathi, English). | Officer Gurpreet completed handover verification in Punjabi in 18 seconds. |
| **Fix 3: Secret Proof Clarity** | Students accidentally typed public descriptions into the private proof field. | Added explanatory tooltips with a lock icon: 'Secret Proof: Only campus security can view this to verify your ownership.' | 100% of participants entered private proofs (serial numbers, engraving) without confusion. |
| **Fix 4: Vite Import Build Error** | Missing `TranslatedText` component triggered build failure in `NotificationBell.jsx`. | Created unified `TranslatedText.jsx` wrapper component with robust fallback handling. | Vite production build compiled cleanly with 0 errors (`vite build` passed). |